

Suraj Dhakad

☎ +91-7879553211 ✉ dhakadsuraj77@gmail.com 🐙 GitHub 🔗 LinkedIn 🌐 Portfolio

PROFESSIONAL SUMMARY

Backend Developer and AI/ML Engineer experienced in building scalable backend systems and LLM-powered applications. Proficient in Python, FastAPI, LangGraph, and LangChain for developing RAG pipelines and multi-agent workflows, with a strong foundation in Machine Learning and Deep Learning using TensorFlow and Scikit-learn. Adept at designing production-grade architectures with PostgreSQL and Redis, integrating vector search and real-time data processing for real-world AI solutions.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Backend:** FastAPI, REST APIs, WebSockets, Pydantic
- **AI / LLM:** LangChain, LangGraph, LangSmith, RAG, Prompt Engineering, Multi-Agent Systems, OpenAI API, Groq API
- **Machine Learning:** Scikit-learn, TensorFlow, Keras
- **Deep Learning :** ANN, CNN, RNN, LSTM, OpenAI Whisper
- **Databases & Vector Search:** PostgreSQL, MySQL, Redis, FAISS, pgvector
- **Developer Tools:** Docker, Git, GitHub Actions, Postman, Railway, Render, Vercel
- **Authentication:** JWT, OAuth2, RBAC
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn

PROJECTS

AI Video Assistant | *Python, OpenAI Whisper, LangChain, RAG, Streamlit, Sarvam AI* 🐙 GitHub

- Built an AI assistant that processes **YouTube videos and local media files** — transcribing, indexing, and enabling conversational Q&A over video content.
- Implemented **speech-to-text transcription** using OpenAI Whisper and a **RAG-based querying pipeline** for contextually accurate answers from video content.
- Generated AI-powered summaries, action items, key decisions, and timestamped highlights automatically using **LangChain chains**.
- Integrated **Sarvam AI** for multilingual transcription and translation, expanding usability for regional language content.
- Designed a **Streamlit conversational UI** for intuitive real-time interactions with video content — zero prompt-engineering required from end users.

Perplexity 2.0 — AI-Powered Search Engine | *LangGraph, FastAPI, React, PostgreSQL, FAISS, Redis, SSE* 🐙 GitHub

- Built a production-style AI search engine with **multi-step reasoning, real-time web search, and contextual answer generation** using LangGraph-orchestrated multi-agent workflows.
- Designed a full **RAG pipeline** using FAISS vector store for semantic search and grounded, citation-backed responses — reducing hallucination risk significantly.
- Implemented **Server-Sent Events (SSE)** for streaming responses, achieving low-latency output comparable to production AI systems.
- Integrated **Tavily Search API** for live web retrieval and Redis caching to reduce redundant API calls and improve response throughput.
- Engineered scalable **FastAPI backend** with PostgreSQL for persistent chat memory, user sessions, and source tracking.

Real-Time Chat Application | *FastAPI, WebSockets, Redis Pub/Sub, PostgreSQL, JWT, RBAC* 🐙 GitHub

- Developed a full-stack real-time messaging system using **WebSockets** with support for concurrent users and low-latency delivery.
- Implemented **JWT authentication and Role-Based Access Control (RBAC)** for secure, scoped access across user roles.
- Integrated **Redis Pub/Sub** for horizontal scalability — enabling message broadcasting across multiple server instances.
- Designed **normalized PostgreSQL schema** with modular REST APIs for user management, message history, and room operations.

CERTIFICATIONS

- **FastAPI for Machine Learning** — CampusX
- **Complete Agentic AI Bootcamp with LangGraph and LangChain** — Udemy

EDUCATION

B.Tech — Electronics & Telecommunication Engineering
Institute of Engineering & Technology (IET), DAVV Indore

2024 – 2028
Indore, MP